CENG60008: Process Optimization

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1 Topic 1. Introduction to Optimization

1.1 Modelling

Optimization model (mathematical program): A mathematical model that seeks values for decision variables x that minimize (or maximize) an objective function $f(x)$, subject to constraints $g(x) \leq 0$. An optimization model is usually set up as follows:

$$\begin{aligned} \min_{x \in \mathbb{R}^{n_x}} \quad & f(x) \\ \text{s.t.} \quad & h(x) = 0 \\ & g(x) \leq 0 \\ & x^L \leq x \leq x^U \end{aligned}$$

s.t. indicates the constraints that the model is subject to, there are 2 types of constraints:

1. Variable-type constraints: $x_1, x_2 \geq 0$
2. Main constraints: $0.5x_1 + 0.2x_2 \geq 0$

Key components of an optimization model:

1. **Decision Variables**: the variables that an optimizer proposes to search for optima.
2. **Constraints**: pre-defined restrictions that influence the location of optima.
3. **Objective/Criterion functions**: the function(s) to optimize, these define the location of optima (minima/maxima) within a given domain, these can define costs, expenses, time-to-completion etc.

Feasible Solutions: solutions that satisfy all defined constraints. **Optimal Solutions**: a subset of feasible solutions, that also achieve objective function values as good, or better than any other feasible solution.

System Boundaries: the line between values that are taken as settled, and values that are to be decided is called the system boundary. The mathematical program lies within the system boundary, which relies on the modeling and analysis of decision variables, subject to parameters (constraints, objective functions). Both the parameters, and decision variables together determine the results measured as output variables.

Subcategories of models: in general, models, and model tractability relies on two main types of models:

1. **Descriptive models**: models that evaluate fixed decisions via brute-force approaches rather through good choices (e.g. simulations, and numerical searches) (more valid, less tractable)
2. **Prescriptive models**: decisions are informed by analytic inferences. (less valid, more tractable - mathematically informed)

Therefore, by construction descriptive models produce fewer inferences than prescriptive models.

Further subcategories of models: models can be further separated into:

1. **Deterministic**: involve fixed outcomes, with usually more mathematically tractable results (more analytical tools are available for result analysis).
2. **Stochastic**: involves uncertainty at inference via methods like stochastic simulation (Monte Carlo analysis) to estimate the mechanics of the true system distribution when n is large enough

As a result, it can be seen that deterministic models may be an approximation when a system has stochastic variance, with higher tractability than stochastic methods.

1.2 Local and Global Optimality

The feasible set S is the collection of choices for decision variables that satisfy all of the constraints:

$$S := \left\{ x \in \mathbb{R}^{n_x} : g(x) \leq 0, h(x) = 0, x^L \leq x \leq x^U \right\}$$

S is plotted by introducing constraints one-by-one, keeping track of the region satisfying all at the same time.

A **global minimum** x^* is a feasible choice for the decision variables, with an objective function at least as good as any other feasible point:

$$f(x^*) \leq f(x), \forall x \in S$$

The optimization problem may have unique minima, multiple minima, or no minima at all, with an optimum solution being found at the optimum objective value f^*.

When defining local optimality, we must check how the local region changes around the proposed optimal vector. A proof of this is as follows:

$$\exists \delta > 0 \quad \text{such that} \quad f(\underline{\mathbf{x}}^*) \leq f(\underline{\mathbf{x}}), \quad \forall \underline{\mathbf{x}} \in S \quad \text{with} \quad \|\underline{\mathbf{x}} - \underline{\mathbf{x}}^*\| < \delta$$

This implicitly constructs a local hyper-sphere of candidate points centered at the local optima, with a radius of δ (finite quantity). The strong inequality $>$ or $<$ indicates strict (isolated) local optima, conversely $\geq$, or $\leq$ indicates weak local optima.

Heuristic/Approximate Optimum: a feasible solution derived from prescriptive analysis that is not guaranteed to yield an exact optimum.

1.3 Tractability, and Feasibility

Tractability: defined as the degree to which the model admits conventional analysis (how feasible analysis is using standard mathematical methods).

The feasible set S can be deemed infeasible due to the following reasons:

1. The feasible set is empty: $S = \emptyset$ (constraints violate all throughout)
2. Set bounds are improperly defined with unconstrained sets, it is computationally intractable for the global minima to be identified within infinite bounds.

1.4 Graphing models

When plotting, take into account the following information:

1. **Contour lines**: draw dotted contour lines of the specified objective function $f(\underline{\mathbf{x}})$ to indicate the level at a given input vector $\underline{\mathbf{x}}^*$.
2. **Constraint boundaries**: these define the region of feasibility, showing where optimisation candidates are allowed to be proposed. When drawing constraint boundaries, constraints can be described as either **active**, meaning that the specific boundary limit directly influences the optima, and **inactive**, meaning the opposite.
3. **Optimum solution vector**: show the exact location of the optimum solution vector $\underline{\mathbf{x}}^*$, found within the feasible region.

1.5 Convexity and global optimality

A function $f : \mathbb{R}^n \to \mathbb{R}$ is said to be:

1. **Convex on S if**: the chord connecting $f(\underline{\mathbf{x}}_1)$ and $f(\underline{\mathbf{x}}_2)$ at any two points $\underline{\mathbf{x}}_1 \neq \underline{\mathbf{x}}_2$ lies above the function between $\underline{\mathbf{x}}_1$ and $\underline{\mathbf{x}}_2$, $f(\gamma\underline{\mathbf{x}}_1 + (1-\gamma)\underline{\mathbf{x}}_2) \leq \gamma f(\underline{\mathbf{x}}_1) + (1-\gamma) f(\underline{\mathbf{x}}_2)$, $\forall\gamma \in [0,1]$
2. **Strictly convex on S if**: $f(\gamma\underline{\mathbf{x}}_1 + (1-\gamma)\underline{\mathbf{x}}_2) < \gamma f(\underline{\mathbf{x}}_1) + (1-\gamma) f(\underline{\mathbf{x}}_2)$, $\forall\gamma \in [0,1]$

A symmetric matrix is said to be

1. **Positive definite** $\lambda_{1,2} > 0$: if all of its eigenvalues are positive.
2. **Positive semi-definite** $\lambda_{1,2} \geq 0$: if all of its eigenvalues are non-negative.
3. **Otherwise, indefinite** $\lambda_{1,2} < 0$, or $\lambda_1 > 0$, and $\lambda_2 < 0$ (saddle point).

where $\lambda_{1,2}$ are the first and second eigenvalues from the eigenvalue decomposition of the hessian.

In situations where the function f is composed of a linear sum of n univariate functions (functions that only depend on 1 variable), then the convexity depends on the individual convexities of each univariate e.g. $f(x_1, x_2) = \frac{10}{\sqrt{x_1}} - \log(x_2) \forall x_1, x_2 > 0$. Given that $\frac{1}{\sqrt{x}}$ is convex (as 2nd oder derivative is positive), and $\log(z)$ is concave (since its 2nd order derivative is negative), and log is negated, becoming convex then the sum of two convex functions leads to a convex function.

2 Linear Programming

2.1 Linear Programs

An optimization model is considered a **linear program** if it has continuous variables, a single linear objective function $f(x) = \underline{\mathbf{w}}^T \cdot \underline{\mathbf{x}}$ (weighted sums of each decision variable x_i), and linear equality/inequality constraints e.g. $\underline{\mathbf{w}}^T \cdot \underline{\mathbf{x}} \leq b$. LPs have several fundamental properties:

1. **Global optimality is guaranteed**: every local optimum is also a global optimum.
2. **Optimal solutions and extreme points**: unique optimal solutions are always found at extreme points, wheras if an LP has multiple optimum solutions, at least one will occur at the **extreme point** (defined in the domain $\mathbb{R}^{n_x}$, where at least n independent constraints are only active at that point, or more simply at the vertices of the constraint envelope bounding the interior of the LP feasible region).
3. **Unboundedness and non-constant objective functions**: if an LP has a non-constant objective function (meaning that $f(x) \neq c$), then the optimal point will lie outside of the interior of the constraint envelope (assuming an unbounded LP, we can see $f(x)$ as an n-dimensional hyperplane with unconstrained global optima at ∞ if maximizing, and $-\infty$ if minimizing).

There are several general types of LP models:

1. **Allocation models**: involves the splitting/allocation of a decision variable bounded by n finite equality constraints, denoted as $g_i(x)$ (e.g. the target number of barrels of oil to be produced).
2. **Blending models**: involves the merging n decision variables to best fulfil the target constraints (e.g. fuel blends for optimum production from n refineries, or scrap blending to manufacture high-grade steel).
3. **Operation planning models**: involves optimizing decisions for optimum production of n products.

Furthermore, the types of decision variables, and constraints are as follows:

1. **Principal decision variables**: denotes the target values for which optima are to be found (principal allocation variables in allocation models, and principal blending variables in blending).
2. **Non-negativity, and bound restrictions**: ensures the correct operating domain for a given decision variable, usually non-negativity constraints $x_i \geq 0$.
3. **Composition constraints**: constraints that sum over the composition n decision variables, enforcing upper and/or lower limits on the properties of the resulting blend.

Sometimes, programming models be time-phased, where time becomes an index dimension on both constraints, and decision variables. A **time horizon** establishes the range of valid time indicies t to be modelled, these can be either **finite** (fixed) or **infinite** (where the final period is wrapped around to the first).

Non-linear programming models can be linearized, resulting in both mathematically, and computationally tractable models where local optima also describe global optima.

2.2 Linearizing non-linear objectives

There are two main strategies for linearizing non-linear programs:

1. **Minimax**: minimizing the maximum, by introducing an intermediate variable f (acts as a new dim), we can then reformulate $f = f(x)$ to be $\min_{f,x} \{f\}$ s.t. $f \geq f_i(x)$.
2. **Maximin**: maximizing the minimum (vice versa): reformulates $f = f(x)$ to be $\max_{f,x} \{f\}$ s.t. $f \leq f_i(x)$.

These 'squeeze' the intermediate variable f to become linear, trading off computational tractability (more difficult to compute).

2.3 Sensitivity Analysis

Key Notation:

1. **Marginal price**: The rate of change of optimal solution z_* for an elementary variation in the RHS of b_i of a main constraint $\sum_{j=1}^{n} a_{i,j} x_j \leq b_i$.

Perturbations on the ith constraint affects the feasible regions for a linear programming model in the following ways:

1. **Relaxing**: relaxing the constraints either keeps, or improves the optimal value.
2. **Tightening**: relaxing the constraints either keeps, or worsens the optimal value.